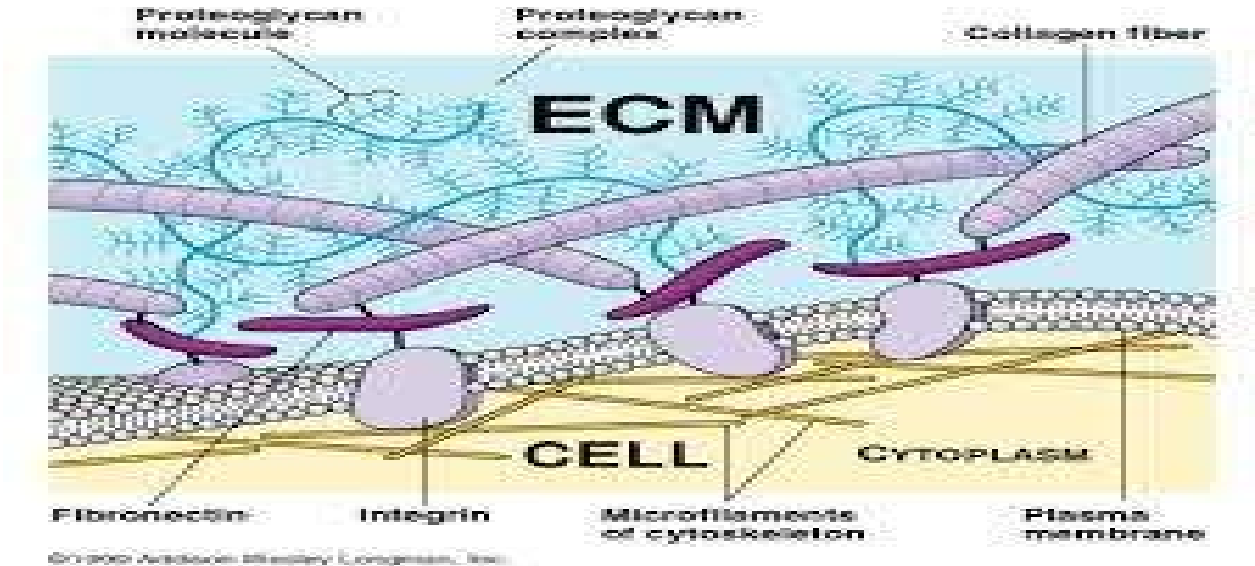
Extracellular Matrix Proteoglycans

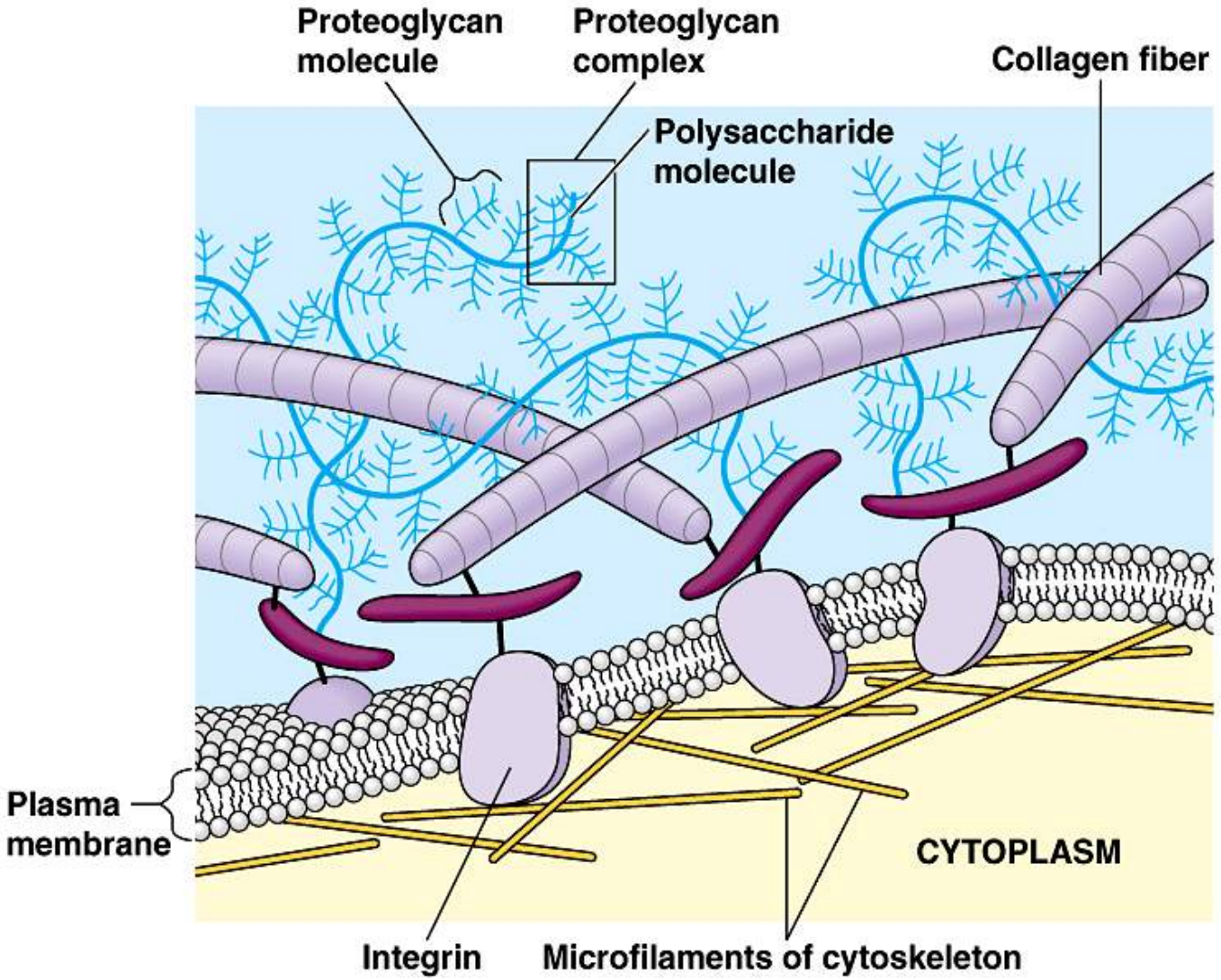


objectives

- Define extracellular matrix
- Define proteoglycans
- Describe the properties, general features of proteoglycans.
- Describe its the synthesis, and degradation.
- Justify their role in contribution to ECM.

ECM: Definitio

- The **extracellular matrix** (ECM) is a collection of **extracellular** molecules secreted by cells that provides structural and biochemical support to the surrounding cells.



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BIOMOLECULES OF ECM

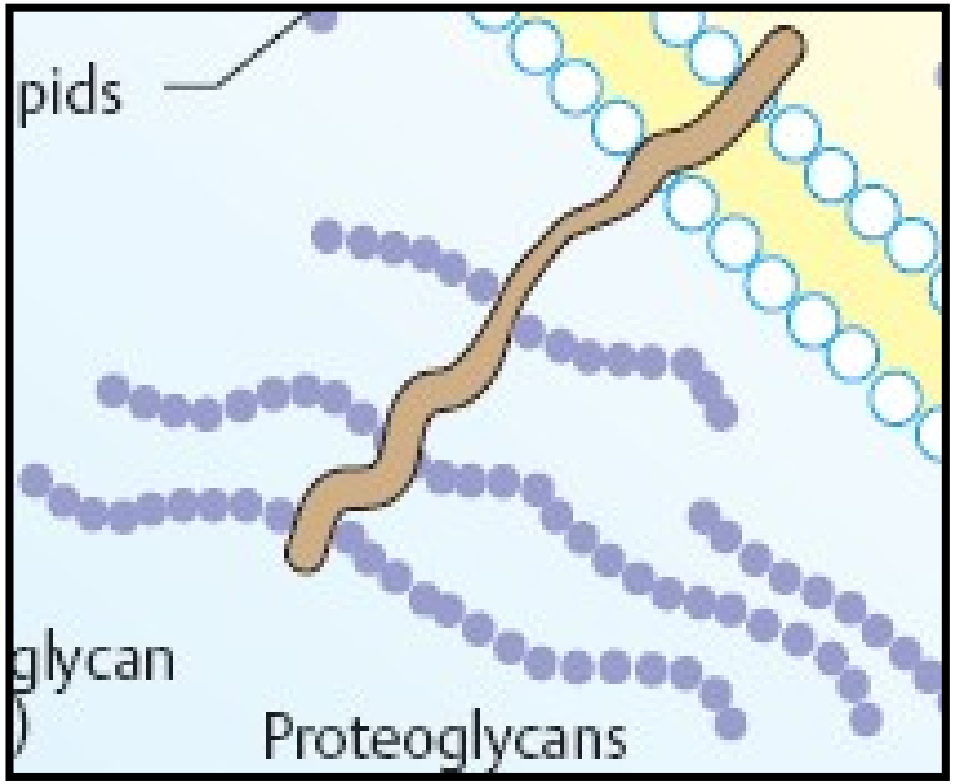
- **Structural Proteins**
- **Collagen** The most abundant protein in the body, making 25%-35% of all the whole-body proteins. contributes to the stability of tissues and organs
  - **Elastin**

    major protein component of tissues that require elasticity such as arteries, lungs, bladder, skin and elastic ligaments and cartilage.

- **Specialized proteins**
- **Fibronectin** cell adhesion, differentiation, anchoring basal laminae to other ECM, blood clothing, link to fibrin
  - **Laminin**
- cell surface receptor, adhesion, differentiation, anchoring the glycoprotein to basal laminae
- **Proteoglycans**
  - GAGs

GLYCOSAMINOGLYCANS

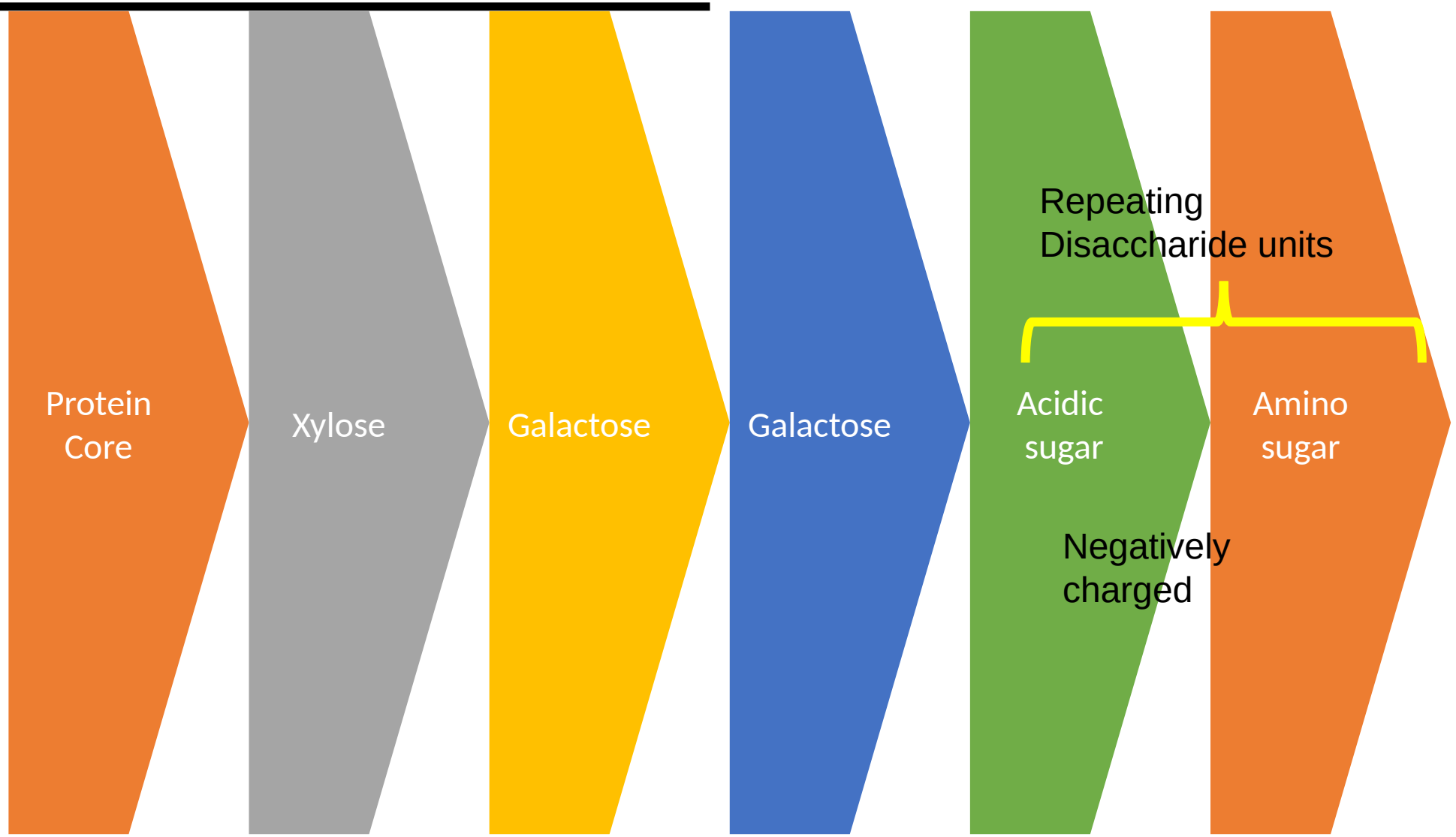
- Present in ECF
- Waterbinding
- Anticoagulant ICF
- List
  - **Hyaluronic acid**
  - **Heparan sulfate**
  - **Heparin**
  - **Keratan sulfate**
  - **Chondroitin Sulfate**



GAG STRUCTURE

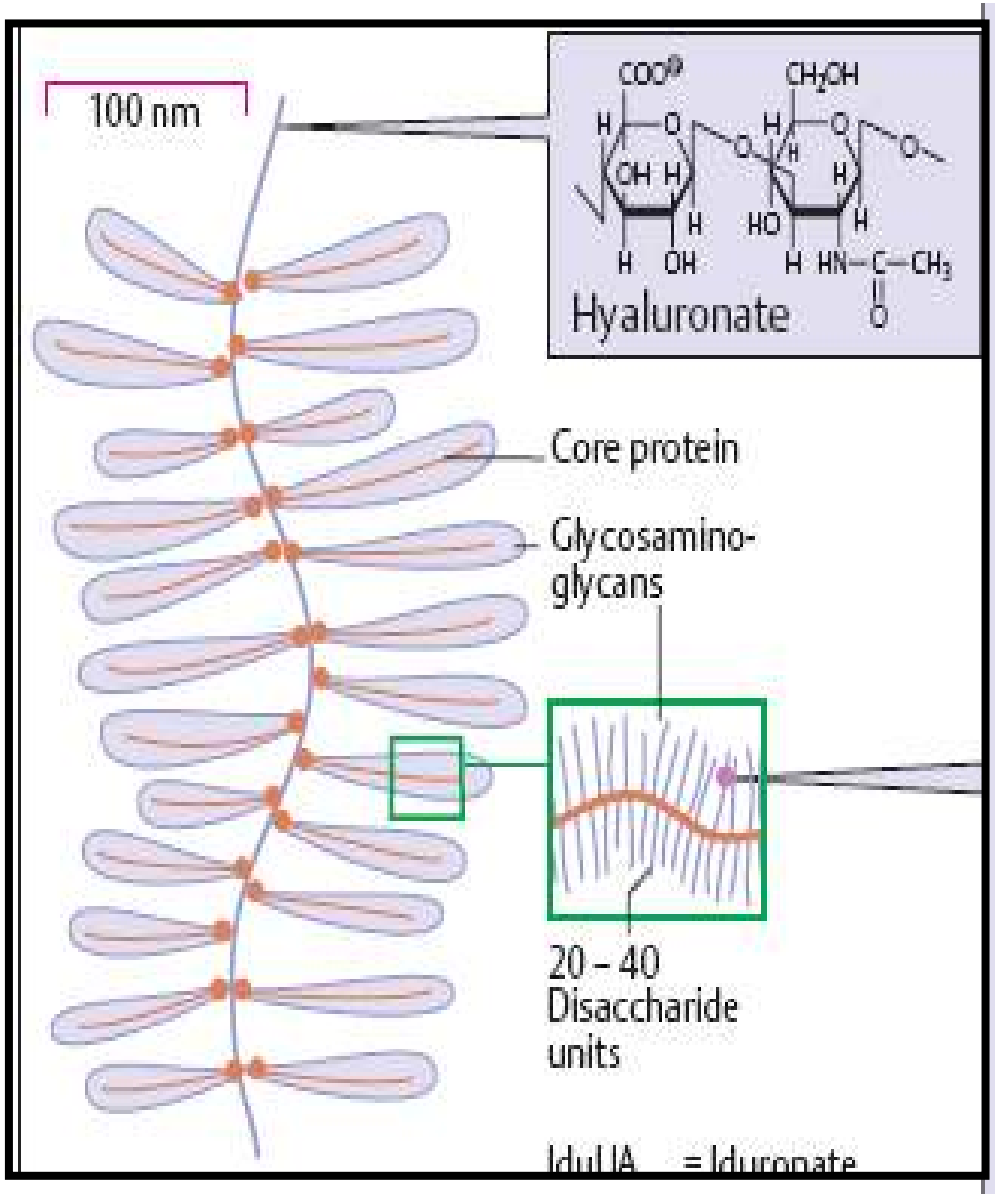
- Protein Core
- OH containing amino acid (serine and threonine)
- Xylose
- Galactose
- Galactose
- Repeating disaccharide units (Acidic sugar-amino sugar)

GAG STRUCTURE



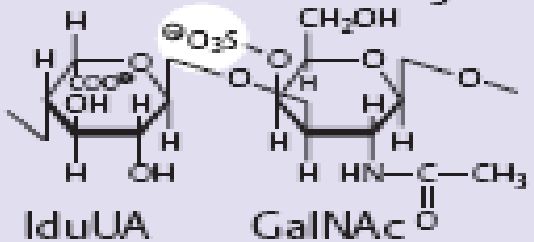
REPEATING DISACCHARIDE UNITS

GAG	Acidic sugar	Amino sugar
Chondroitin Sulfate	GlcUA	GalNAC
Keratan Sulfate	Gal	GlcNAC
Dermatan Sulfate	IdUA	GalNAC
Heparan Sulfate	GLcUA	GlcN
Heparin	GlcUA/IdUA	GlcN
Hyaluronic Acid	GlcUA	GlcNAC

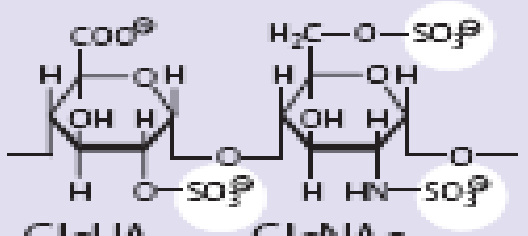


Disaccharide units

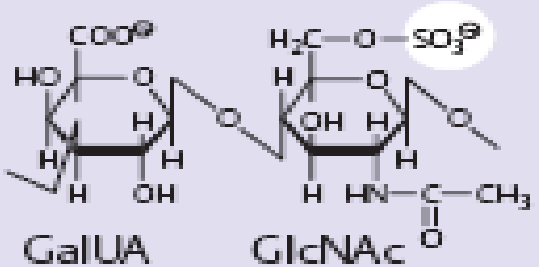
– Uronic acid – Aminosugar –



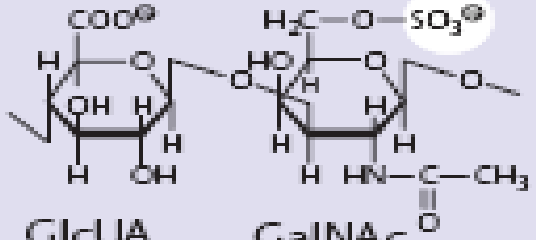
Dermatan sulfate



Heparin



Keratan sulfate

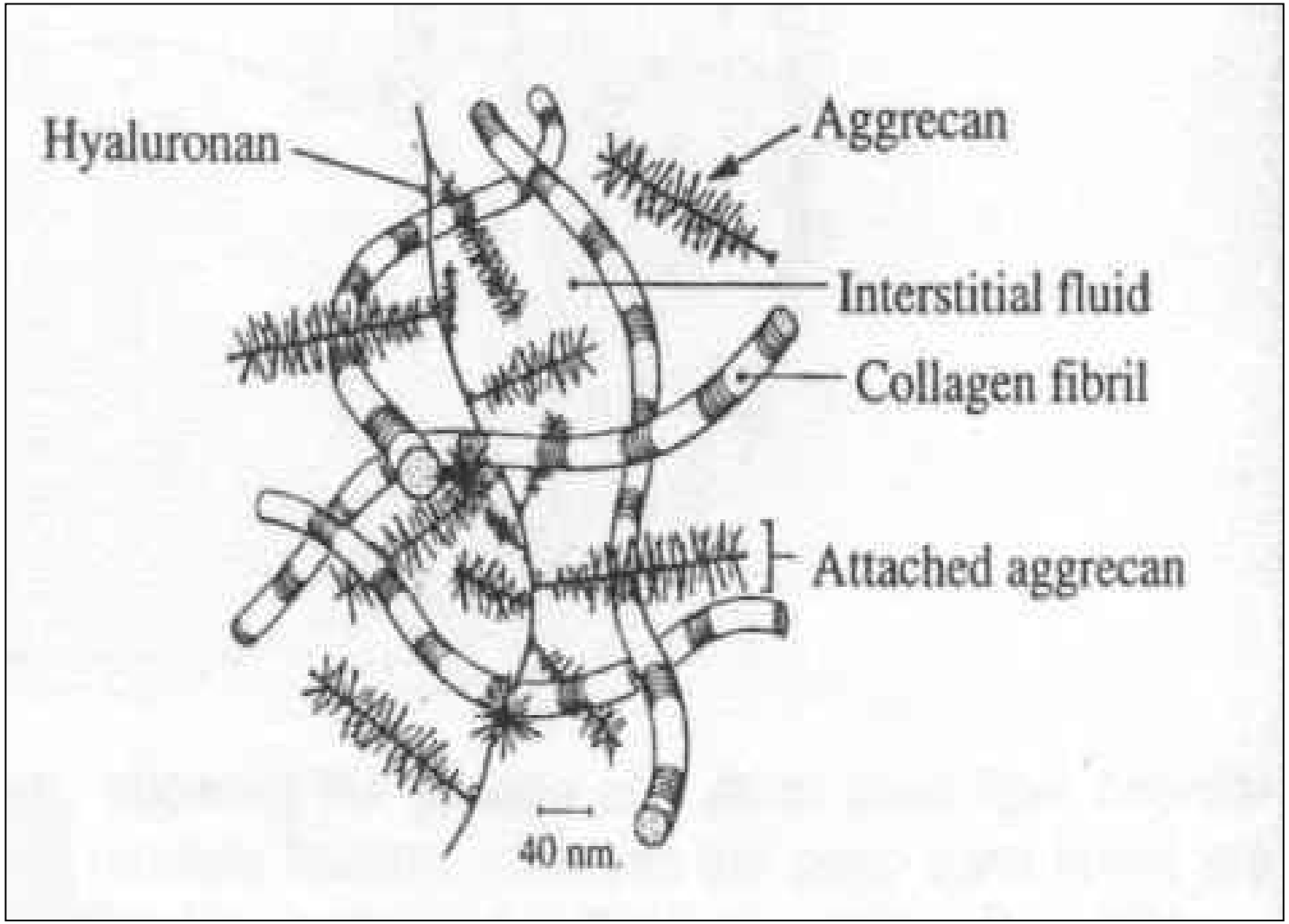


Chondroitin 6-sulfate

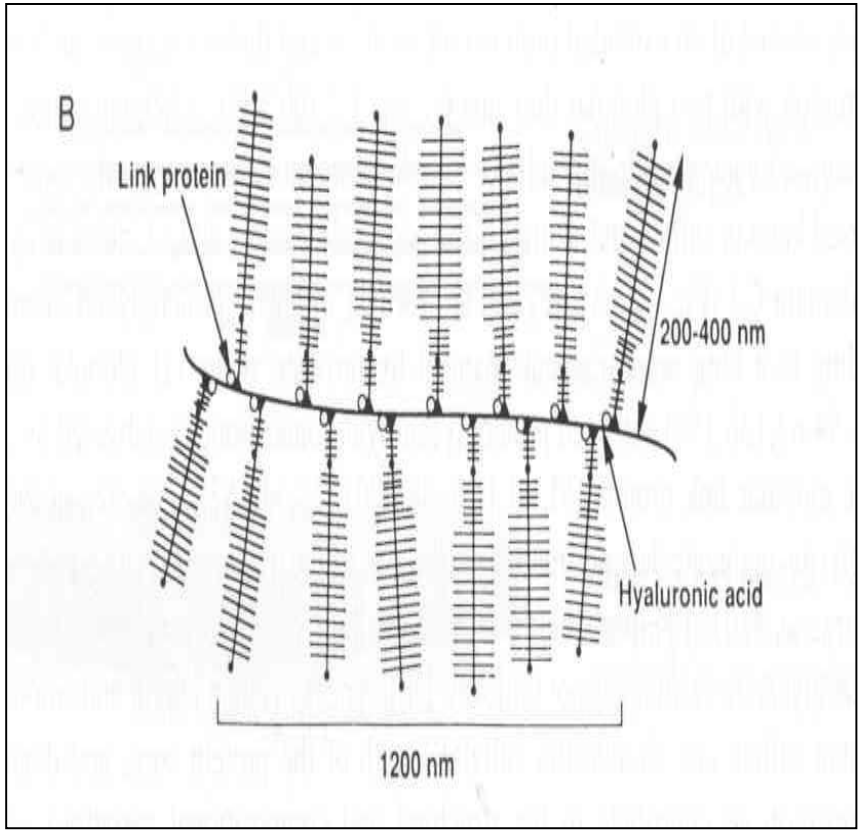
GROUND SUBSTANCE

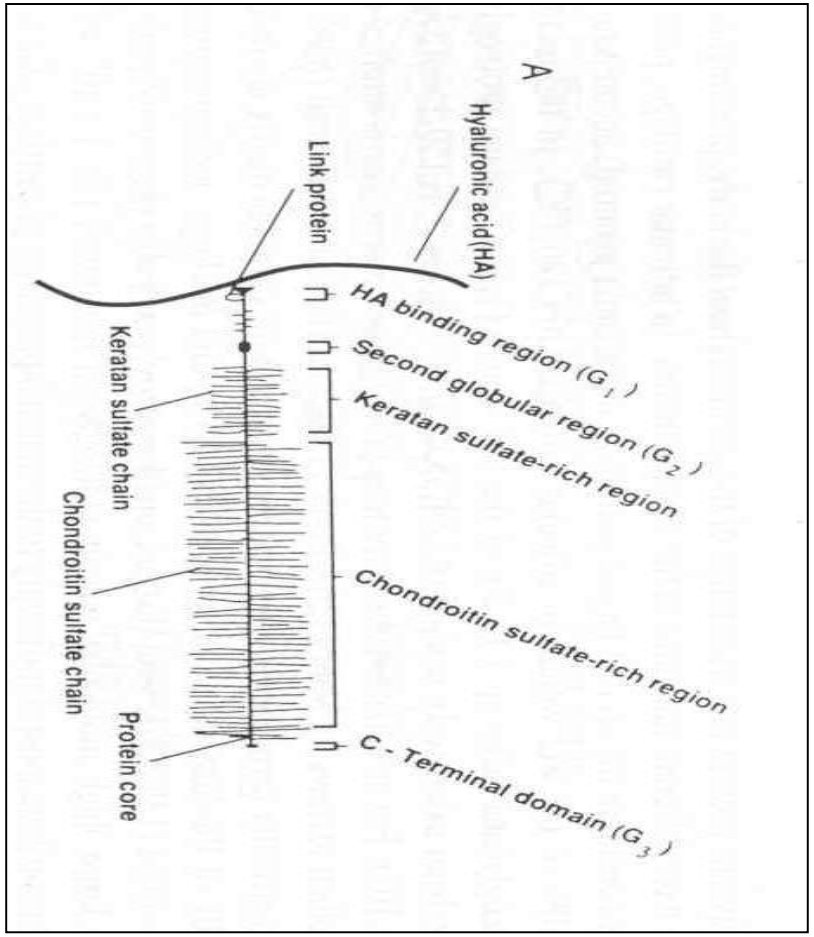
- Highly hydrated complex of mixture of proteoglycans and glycoproteins
- Proteoglycans are linear polysaccharides of repeating disaccharide units composed of **hexosamine & uronic acid**
  - **Glucosamine/Galactosamine**
  - **Glucoronic acid/Iduronic acid**

PROTEOGLYCAN: PROTEIN WITH BOUND SIDE CHAINS (GLYCOSAMINOGLYCANS)



PROTEOGLYCAN: PROTEIN WITH BOUND SIDE CHAINS (GLYCOSAMINOGLYCANS)



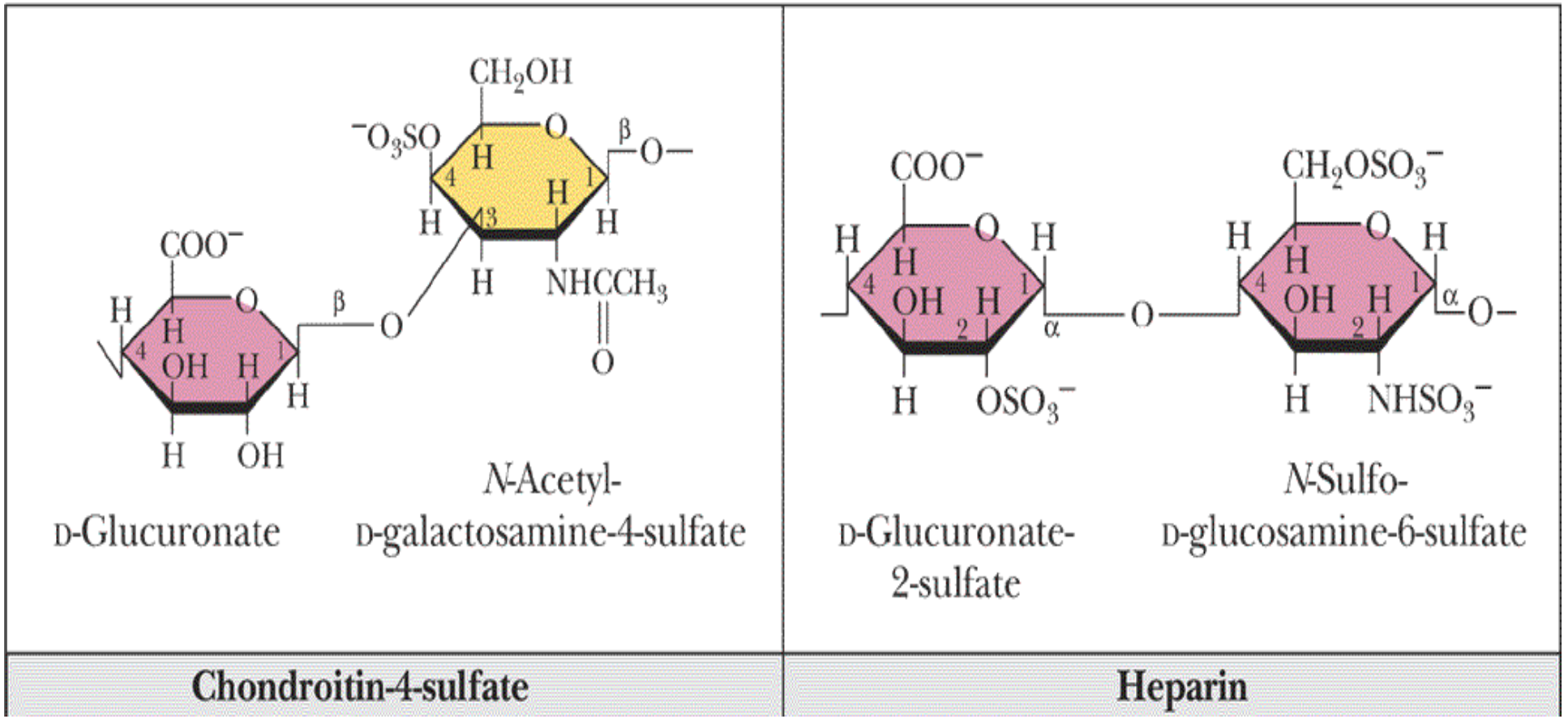


HETERO POLYSACCHARIDES (GAG) OF THE EXTRA-CELLULAR MATRIX

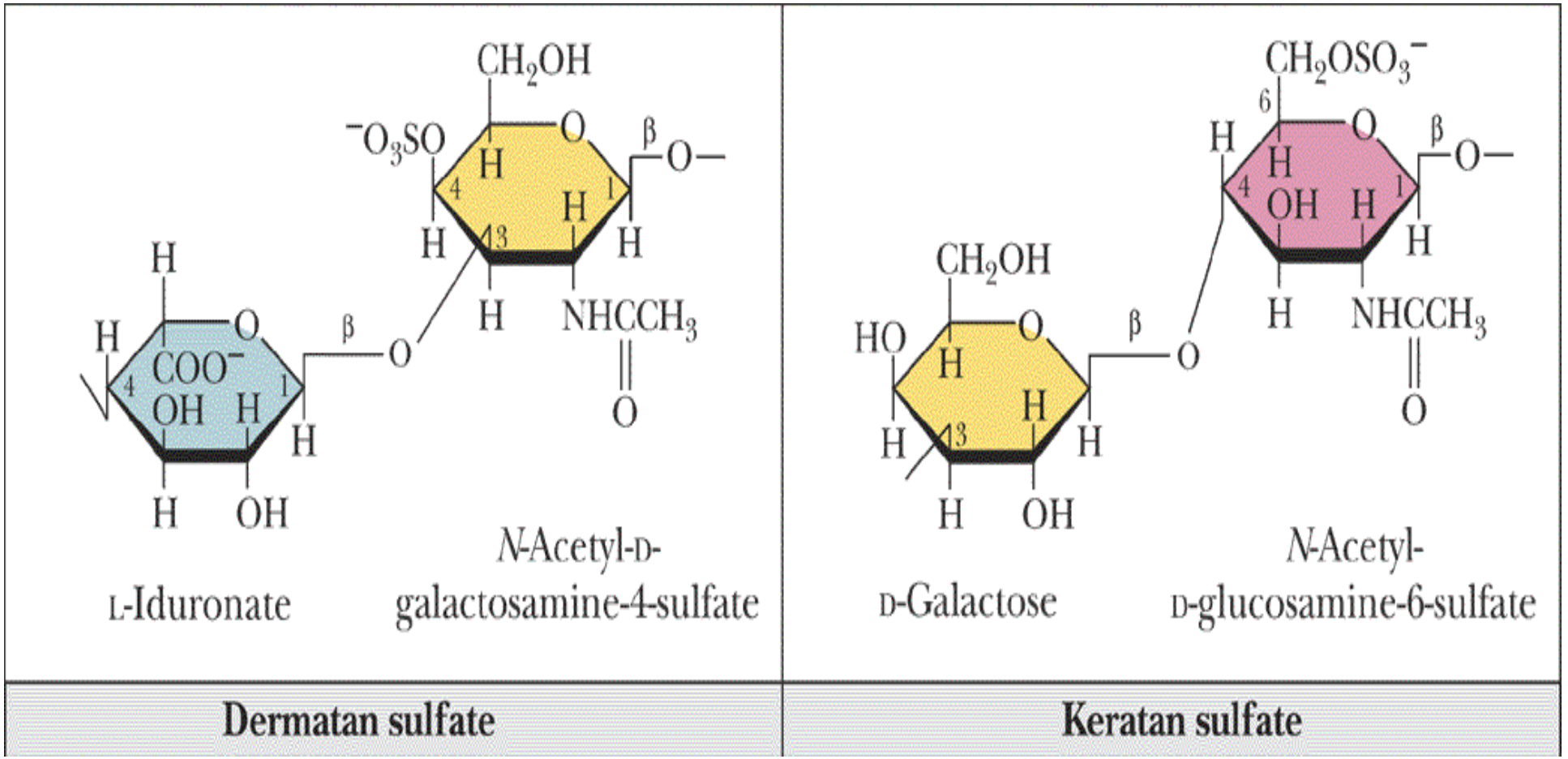
1) **Hyaluronic acid - glassy and translucent**
   - lubricants in joints, **cartilage**, and tendons
   - hyaluronidase in pathogenic bacteria and sperm
2) **Chondroitin sulfate**
   - **cartilage**, tendon, ligament, and walls of the aorta
3) **Dermatan sulfate**
   - skin, blood vessels, and heart valves
4) **Keratan sulfate**
   - cornea, bone, **cartilage**, horn, hair, hoofs, nails and claws
5) **Heparin - natural anticoagulant made in mast cells**
   - bind antithrombin, then bind and inhibit thrombin

GLYCOSAMINOGLYCANS

Involved in a variety of extracellular functions; chondroitin is found in tendons, cartilage and other connective tissues



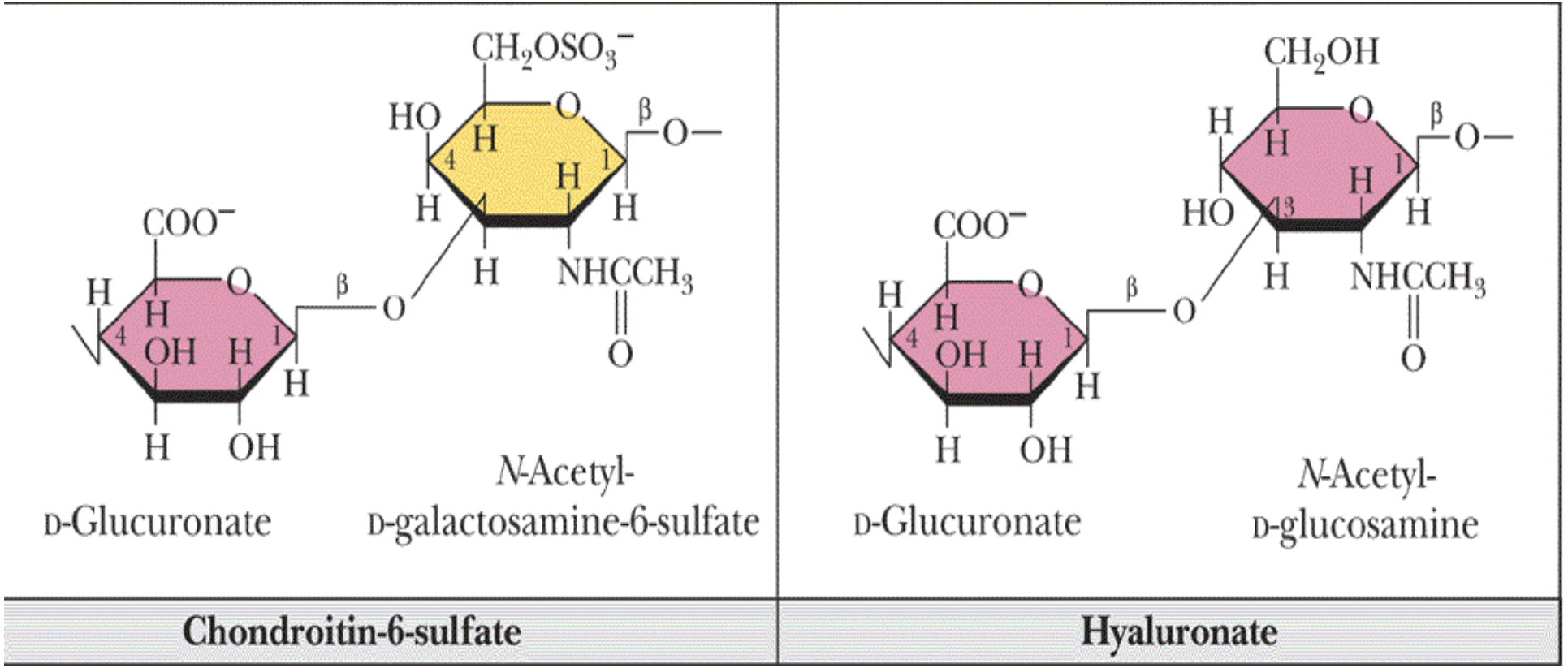
GLYCOSAMINOGLYCANS

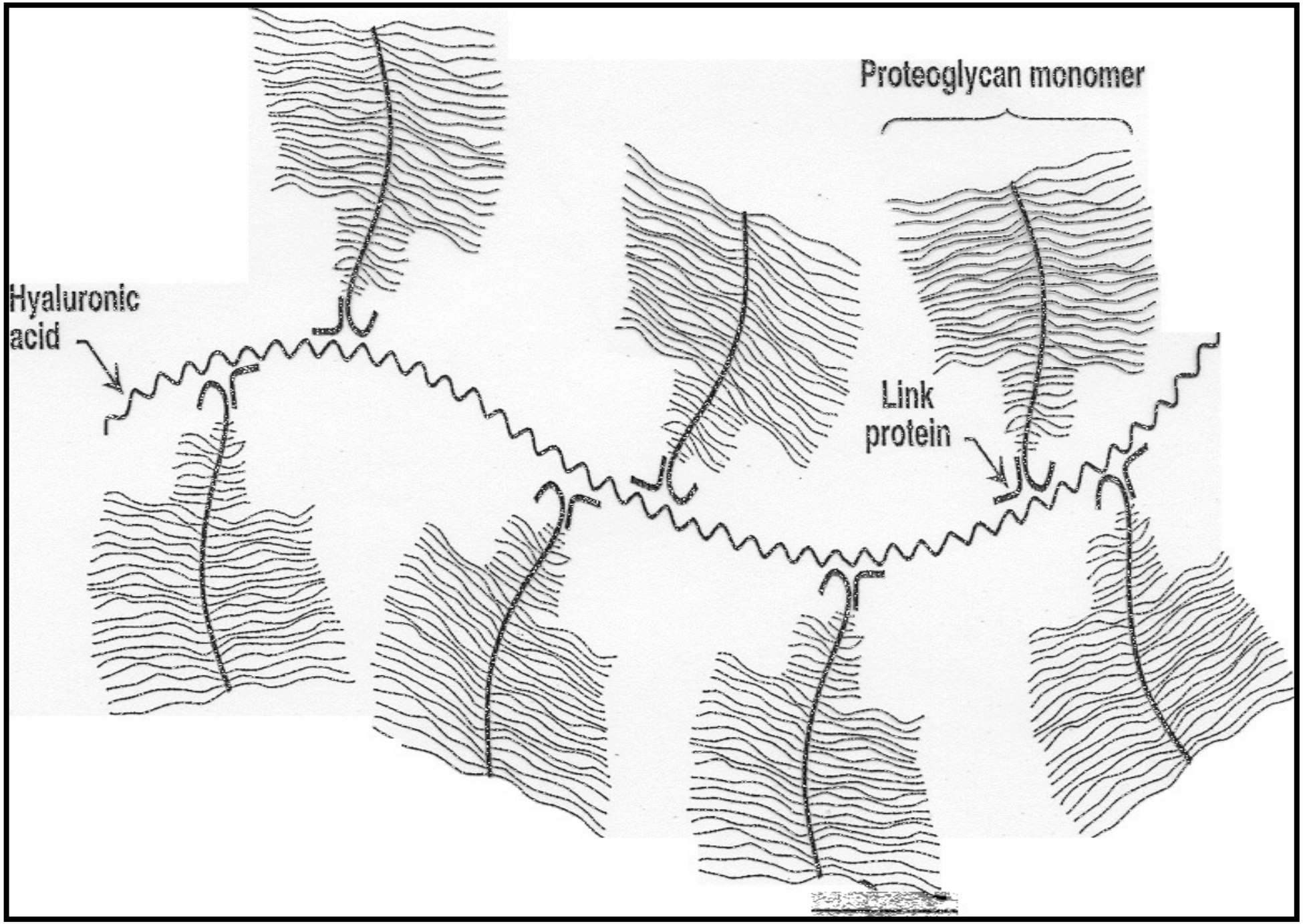


Dermatan sulfate	Keratan sulfate

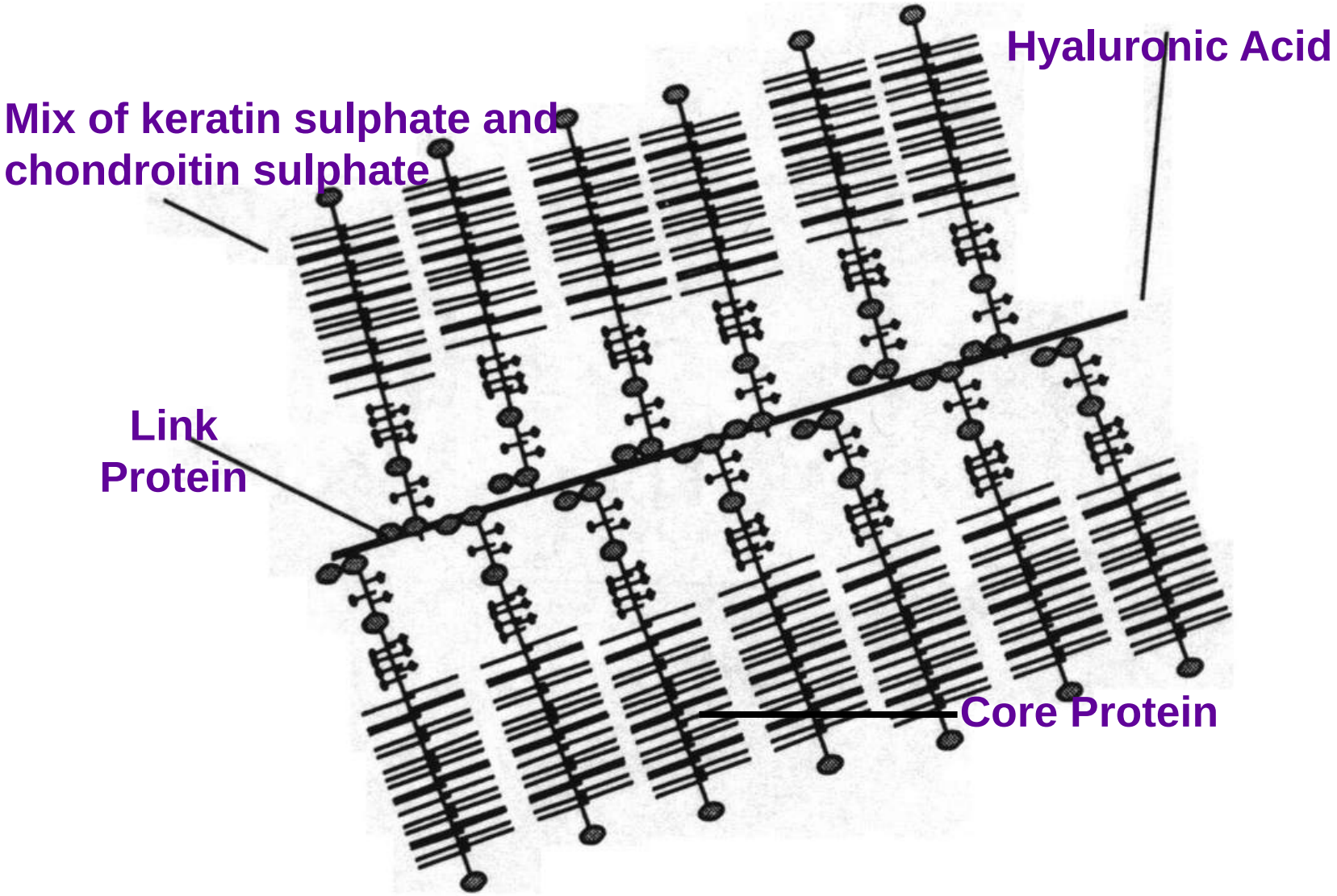
GLYCOSAMINOGLYCANS

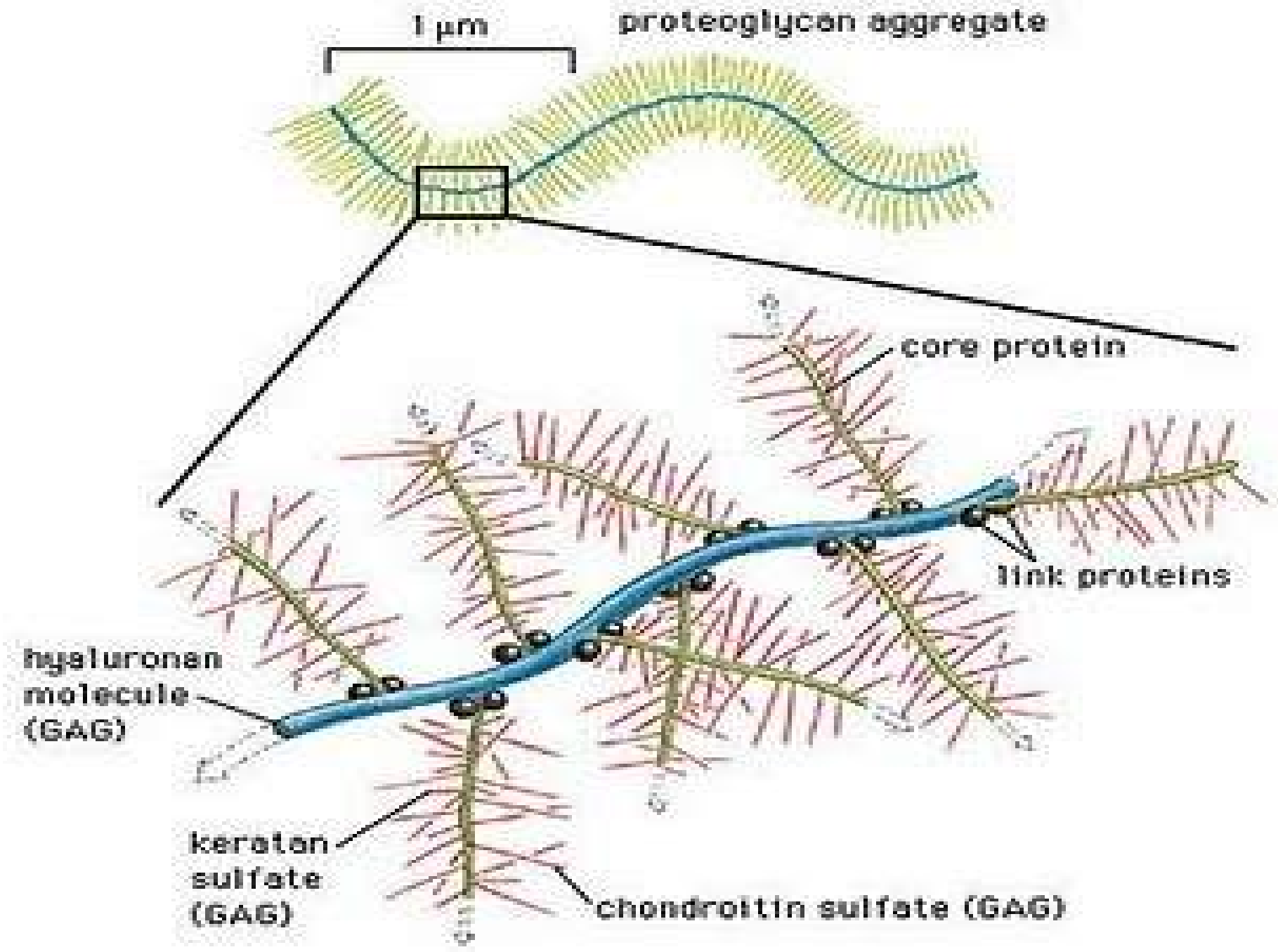
A characteristic of glycosaminoglycans is the presence of acidic functionalities (carboxylate and/or sulfates)




Proteoglycan monomer
Hyaluronic acid
Link protein

Proteoglycan Aggregate From Cartilage




1 µm
proteoglycan aggregate
core protein
link proteins
hyaluronan molecule (GAG)
keratan sulfate (GAG)
chondroitin sulfate (GAG)

GLYCOSAMINOGLYCANS FUNCTIONS

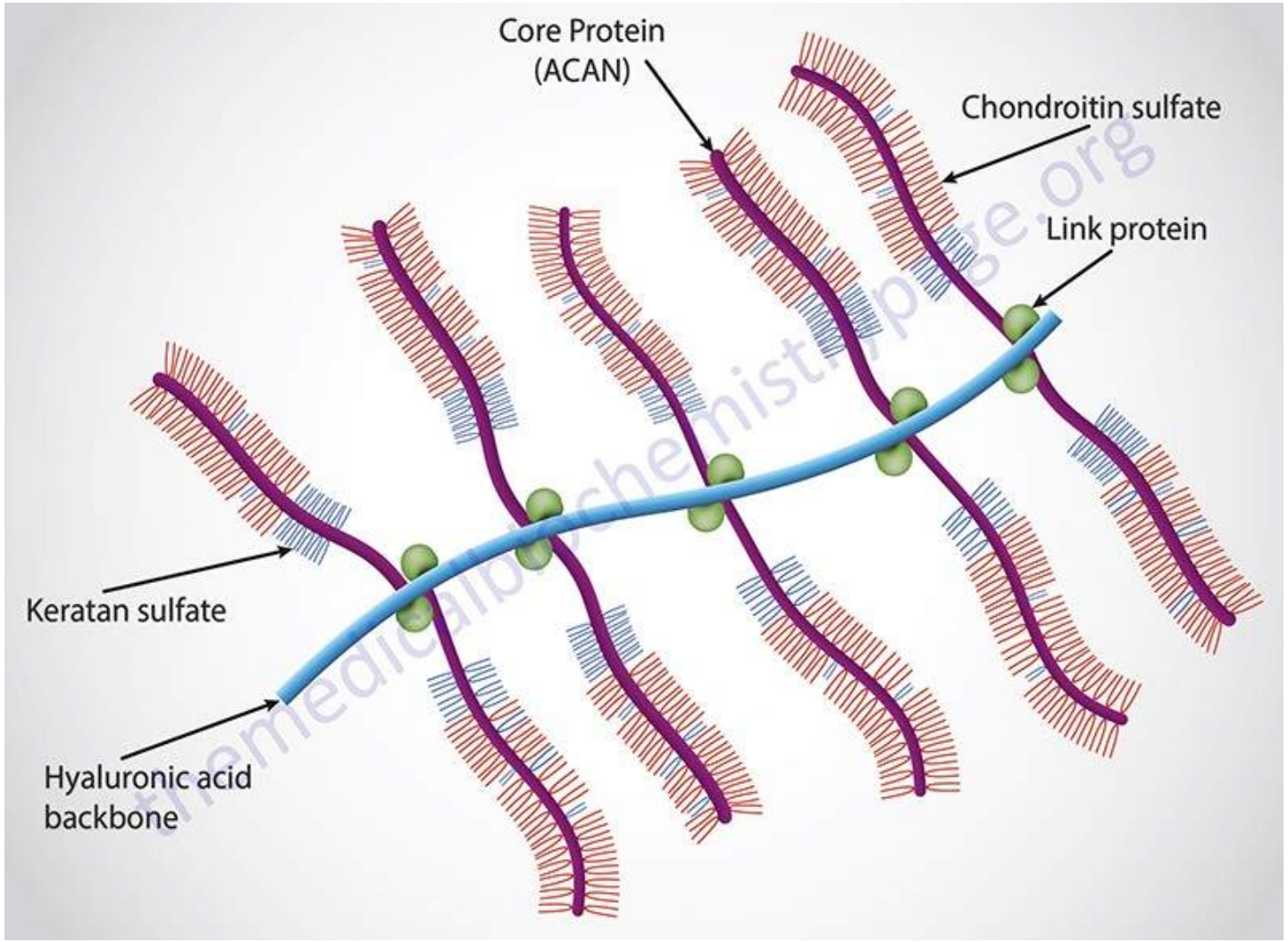
- Compressive strength
- Negatively charged therefore
  - Attract positively charged ions
  - water

GLYCOSAMINOGLYCANS ABSENCE

- **Cause**
  - trauma or osteoarthritis,
- **Result**
  - the mechanical stiffness of the tissue is dramatically reduced, and the functional integrity of the cartilage is compromised.

Proteoglycan: Definition

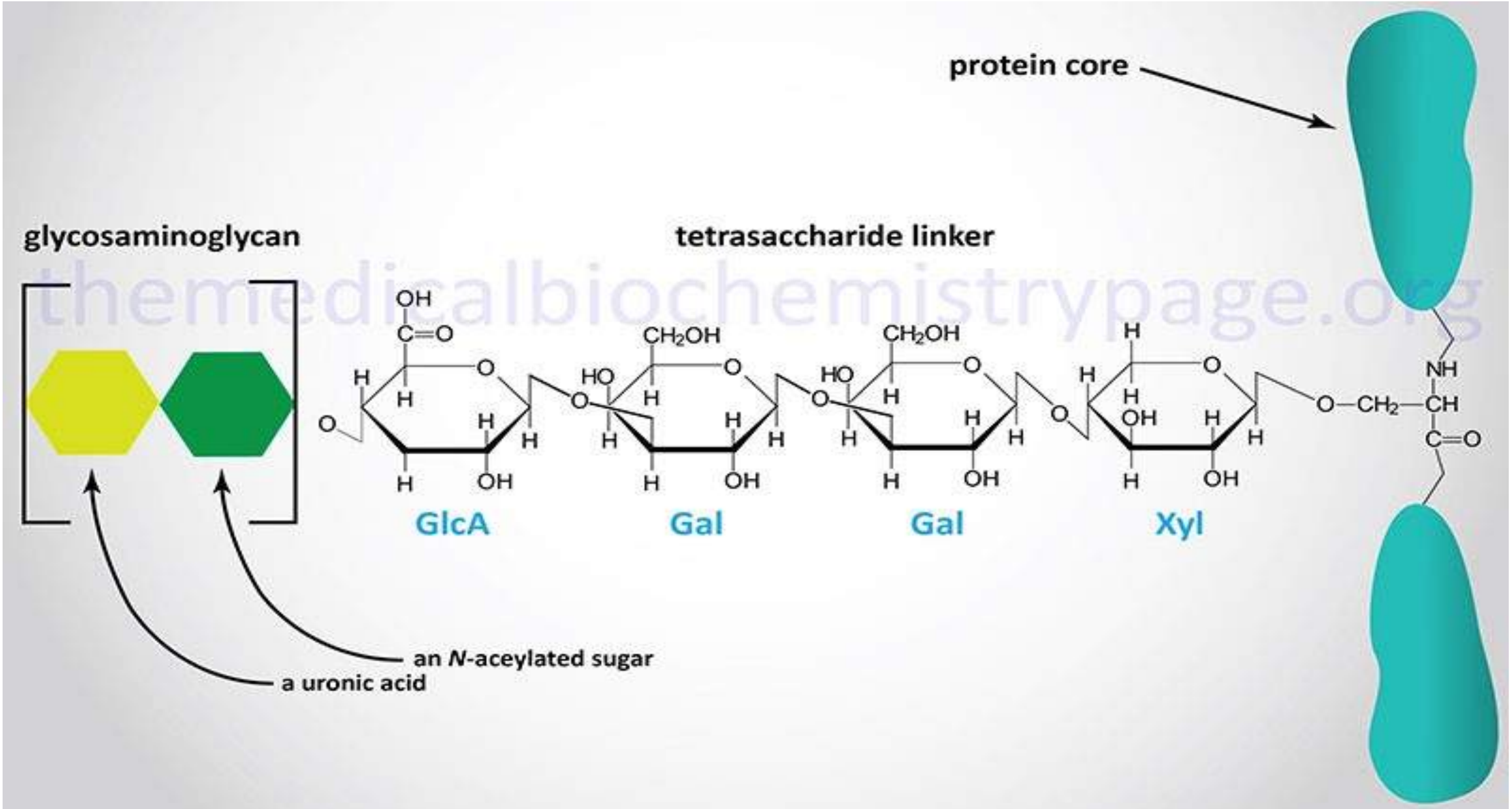
- **Proteoglycans** are proteins that are heavily glycosylated. The basic **proteoglycan** unit consists of a "core protein" with one or more covalently attached glycosaminoglycan (GAG) chain(s).



Structure and Properties:

- The majority of GAGs in the body are linked to core proteins, forming proteoglycans (also called **mucopolysaccharides**).
- The GAGs extend perpendicularly from the core in a brush-like structure.
- The linkage of GAGs to the protein core, in most but not all proteoglycans, involves a specific tetrasaccharide linker composed of a glucuronic acid (GlcA) residue, two galactose (Gal) residues, and a xylose (Xyl) residue forming a structure

- $GAG_{(n)}$–GlcA–Gal–Gal–Xyl–Ser–protein.



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PROTEOGLYCAN STRUCTURE

- The chains are linear (unbranched)
- They are linked to the protein core via a serine or threonine (O-linked) (except HA) forming ***Proteoglycan***.
- Negative charge due to OH^-, COO^-, and SO_4^-,
- PG are hydrophilic and act as poly anion attracts +ions (K and Na)
- Highly hydrated and viscous.

PROTEOGLYCAN FUNCTIONS

- **Modulation of cell growth processes**
  - Binding of growth factor proteins by proteoglycans in the glycocalyx provides a reservoir of growth factors at the cell surface.
- **Cushioning in joints**
  - Cartilage matrix proteoglycans absorb large amounts of water. During joint movement, cartilage is compressed, expelling water!

REFERENCES

Harper's Biochemistry

29th edition

Chapter 48